

IKHLAQ ZAHIR

Computer Science Graduate, Game Development, HCI & AI-Driven Interactive Systems

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Google Play (Developer):

play.google.com/store/apps/developer?id=IsolatedVertex+Studios



PROFESSIONAL SUMMARY

Computer Science graduate with hands-on, end-to-end experience independently designing, developing, testing, and deploying games and Android applications. Development background spans Unity/C#, Kotlin/Android, algorithm design, procedural content generation. Seeking graduate research opportunities combining a software and game-development background with formal research in game technology, HCI, AI, interactive systems, or educational games.

CORE COMPETENCIES

Game Development: Unity Engine, C#, 2D/3D Development, Game Mechanics & Systems Design, Character Controllers, AI-Driven Gameplay, Pathfinding & Navigation, Animation Systems, Physics-Based Interactions, Game Architecture, Procedural Content Generation, Constraint-Based Puzzle Design

Android Development: Kotlin, Android SDK, Jetpack Compose, Android UI/UX, SQLite, JSON Processing, REST/API Integration, In-App Billing, Google Play Deployment, Mobile Application Architecture

AI & Machine Learning: Python, PyTorch, Machine Learning Fundamentals, Generative AI, Large Language Models (LLMs), Transformer-Based Models, AI-Assisted Development, On-Device/Lightweight AI

HCI & Interactive Systems: User Interface Design, UX-Oriented Development, Interactive System Design, Player Experience, Human-Computer Interaction, Interactive Learning Systems, Game-Based Learning, User-Centered Design

Tools & Platforms: Visual Studio, Android Studio, Unity, Google Colab, Hugging Face, Gradle, JSON, SQLite

EDUCATION

Bachelor of Science in Computer Science

October 2024

University of Management and Technology (UMT), Lahore, Pakistan | 2020 – 2024

Intermediate in Pre – Engineering

2019

Aga Khan Higher Secondary School, Gilgit

Matriculation (Science)

2017

Aga Khan Higher Secondary School, Gahkuch

SELECTED PROJECTS

Wordle World

December 2025

Status: Open Testing

Android · Kotlin · Multiplayer · Narrative Game Systems

- Combined logic puzzle and narrative thrill to bring a balanced game
- Designed realm-based difficulty progression and a four-part power-up system.
- Integrated monetization and Google Play deployment systems.

Research Relevance: Multiplayer Game Systems, Narrative Design, Mobile Game Development, Interactive UI.

Numatrix

February 2026

Status: Open Testing

Android · Kotlin · Algorithmic Puzzle Design

- Designed puzzle rules, constraint logic, and difficulty scaling from 5×5 to 10×10 grids.
- Incorporated procedural content generation and created an efficient way to generate complex levels efficiently across different grid sizes
- Implemented dynamic grid interactions and validation/error-state systems.

Research Relevance: Procedural Content Generation, Computational Game Design, Constraint Solving, Algorithmic Gameplay.

MemWord

March 2026

Status: Closed Testing

Android · Kotlin · Educational Application

A vocabulary-retention app in closed testing on Google Play, built on a four-step multi-sensory learning model (Observe, Listen, Read, Write) with speech-recognition-based pronunciation practice.

- Designed the four-step "Neural Anchor" learning process and its data model.
- Implemented speech-recognition practice with homophone-aware answer checking.
- Built the Ultimate Test and Knowledge Tower retention systems, with a daily five-word pacing system for spaced, sustainable learning.

Research Relevance: Educational Technology, HCI, Game-Based/Interactive Learning.

WordLock

In Development · May 2026

Procedural Word Puzzle Generation · Kotlin · Algorithms

A procedural word puzzle game based on simultaneously satisfying row and column word constraints.

- Designed algorithms for generating valid puzzle grids from a dictionary, ensuring each letter is valid across its corresponding row and column.
- Developed constraint-based word placement and validation logic; implemented dictionary preprocessing and word indexing by length.
- Investigated candidate pruning and search strategies; explored difficulty scaling through grid size and permitted word lengths.

Research Relevance: Procedural Content Generation, Constraint Solving, Algorithmic Game Design, Complex Computation.

Scientific Civilization

In Development · July 2026

2D Educational/Simulation Game · Pre-Production / Active Development

A large-scale 2D game centered on reconstructing scientific and technological civilization from fundamental resources and discoveries. Players progressively develop scientific knowledge, manufacturing processes, and increasingly sophisticated technologies, guided by a technology-dependency model in which early experiments become prerequisites for later inventions.

- Designing resource extraction, material processing, and manufacturing systems.
- Building a technology-dependency model distinct from conventional crafting recipes, with a knowledge-progression system tied to in-game scientific experiments.
- Developing character-driven narrative progression alongside the technology tree.

Research Relevance: Educational Games, Serious Games, Game-Based Learning, HCI, Interactive Learning, Simulation, Systems Design. Research direction: how complex scientific and technological knowledge can be represented through interactive systems and communicated through gameplay.

HSK Vocabulary

March 2026

Status: Closed Testing

Android · Kotlin · Educational Application

A vocabulary-learning application structured around flashcards, testing, progress tracking, contextual examples, pinyin, and mnemonic learning.

- Designed interactive flashcard and testing systems.
- Implemented progress tracking and animated learning interfaces.
- Structured vocabulary data for efficient mobile access.

Research Relevance: Educational Technology, Interactive Learning, HCI, Mobile Learning.

PROFESSIONAL EXPERIENCE

Independent Game & Application Developer

2024 – Present

Self-Directed · Concept to Google Play Deployment

Independently designed and developed multiple games and Android applications from concept through implementation, testing, iteration, and Google Play deployment. Several projects have been submitted to Google Play testing tracks, providing practical experience with the complete application development lifecycle.

- Game and application architecture; gameplay and interaction design.
- Programming, debugging, and algorithm development.
- UI/UX implementation and procedural generation.
- AI and gameplay systems; Android performance optimization.
- Database and data management.
- Testing, iteration, and Google Play release management.

Tech Growth & Sales Lead

February 2020 – Present

Mavericks United / Devsinc Group

Experience across technology-focused business development and technical sales, including B2B/B2C technology engagement, technical requirement analysis, lead generation and qualification, CRM management, and coordination between technical and business stakeholders. This role has strengthened requirements analysis, communication, and the ability to understand technical solutions from both engineering and user perspectives.

RESEARCH INTERESTS

- Game Development, Game Architecture, and Game Technology
- Game AI and Intelligent Interactive Systems
- Human-Computer Interaction (HCI) and User Experience
- Procedural Content Generation and Algorithmic Game Systems
- Educational and Serious Games
- Human-AI Interaction
- Interactive Learning Systems
- XR, Immersive Interaction, and Interactive Experiences
- Computational Game Design and Game Architecture

SELECTED RESEARCH DIRECTIONS FOR GRADUATE STUDY

- Game-Based Learning, how interactive game mechanics can communicate complex scientific and technical concepts more effectively than conventional information presentation.
- Intelligent Game Systems, how AI can adapt game experiences according to player behavior, knowledge, engagement, and ability.
- Procedural Content Generation, how algorithms can generate meaningful game content while maintaining constraints, coherence, difficulty, and player experience.
- Human-AI Interaction in Games, how AI agents can become interactive teachers, companions, NPCs, or collaborators within game environments.
- Interactive Simulation, how games can model complex real-world systems while allowing players to learn through experimentation and discovery.
- HCI and Immersive Experiences, how novel interaction technologies, XR, and interactive environments can create more engaging and effective learning and gaming experiences.

RESEARCH SKILLS

- Independent problem solving and algorithmic thinking.
- Technical prototyping and system design.
- Experimental thinking and iterative development.
- Translating concepts into working prototypes.

- Cross-disciplinary thinking spanning computing, games, AI, and HCI.

ADDITIONAL INFORMATION

Programming Languages: C#, Kotlin, Python, JavaScript/TypeScript

Game Engines: Unity, Android Studio

Mobile Platform: Android

AI/ML Tools: PyTorch, Hugging Face, Google Colab

Languages Spoken: English, Urdu

Location: Pakistan